STONet-S*: A Knowledge-Distilled Approach for Semantic Segmentation in Remote Sensing Images

Wujie Zhou, *Senior Member, IEEE*, Penghan Yang, Weiwei Qiu, and Fangfang Qiang

***Abstract*— Semantic segmentation of remote sensing images is a critical research domain. The integration of cross-modal features enhances stability in intricate environments. Despite the impressive performance of existing methods, their complexity and parameter demands remain significant. Our proposed STONet-S*, a stepped transmission optimization network (STONet) with knowledge distillation (KD), extracts insights from a pretrained extensive teacher network and transfers them to an untrained compact student network. Initially, a group enhancement and interaction unit (GEIU) correct for background noise influence and seamlessly integrates cross-modal features. Additionally, we introduce a stepped transmission decoder (STD) comprising a stepped capture module (SCM) and a self-reverse revision module (SRRM) to capture multiscale information from the ground up. Furthermore, leveraging the frequency domain, we employ frequency-awareness KD using a discrete cosine transform (DCT) and octave convolution to separate high and low-frequency maps, which are subsequently transferred to the student network. Last, detail-delivery and stepped-response KD (SRKD) mechanisms enhance the learning capacity of the student network. Through extensive experimentation on two datasets, STONet-S* demonstrates superior segmentation accuracy by achieving remarkable results with only 7.19 M parameters. The corresponding code repository can be accessed at: https://github.com/MAXHAN22/STONet.**

***Index Terms*— Frequency-awareness knowledge distillation (FAKD), group enhancement and interaction unit (GEIU), remote sensing images (RSIs), semantic segmentation, stepped transmission decoder (STD).**

I. Introduction

REMOTE-sensing images (RSIs) represent long-distance Earth surface imagery acquired from platforms such as airplanes or satellites. Semantic segmentation, a well-established technique in image processing [1], [2], has found extensive application in RSIs [3]. In this approach, each RSI pixel is assigned a predefined category that reflects its semantic content. These categorized pixels, represented in distinct colors, collectively form a segmentation map [4]. As urban areas evolve and modernize, the semantic information embedded in RSIs becomes crucial for informed decision-making. Consequently, semantic segmentation of RSIs has gained prominence in resource planning [5], geological disaster prevention [6], and urban planning [7].

Over the last several years, significant progress in deep learning and computer technology has propelled convolutional neural networks (CNNs) to the forefront. These networks autonomously acquire features from images and extract crucial information for semantic segmentation [8], [9], [10]. Recent research indicates that incorporating an additional modality can offer complementary insight, addressing segmentation challenges in complex scenes [11], [12], [13], [14], [15].

Effectively fusing cross-modal information presents a significant challenge. Existing fusion methods [16], [17] that simply add or concatenate multimodal features may render the learned features vulnerable to background interference or overlook details in specific regions, resulting in suboptimal feature representation. To address these issues, prior models have explored incorporating attention mechanisms [18], [19], [20] or gate mechanisms [21] during feature fusion for information filtering. However, they fail to consider that different-scale features require distinct fusion strategies based on their characteristics, without which we cannot obtain satisfactory results.

Following the acquisition of expressive features across various scales, the second challenge lies in their effective aggregation. Without incorporating additional steps to integrate multiscale features directly, low-level features spontaneously introduce interfering information, thereby compromising the performance of the model. While layer-by-layer approaches for aggregating multiscale features [22] offer simplicity and efficiency, they fail to exploit the differences between these features, rendering the aggregated representation susceptible to noise. Moreover, the upsampling process employed in the decoder results in critical information loss, leading to incomplete predictions. Therefore, a suitable decoder with the ability to resist interference and maintain information integrity is urgently needed.

The third challenge associated with models lies in their ability to achieve high segmentation accuracy in RSIs. However, these models often incur substantial computational costs owing to their large model sizes and reliance on extensive training datasets. For instance, Zhang et al. [23] integrated a transformer into a CNN to create a hybrid network that enhances feature extraction and establishes robust spatial dependencies for RSIs. Similarly, He et al. [24] devised a multibranch semantic fusion module using dilated convolutions, effectively enhancing semantic information while significantly increasing

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the parameter count of the model. Additionally, Sun et al. [25] proposed a complex model with a two-layer decoder structure, leveraging supervised features from the main decoder to correct and refine the output of the subdecoder. Despite their remarkable achievements in RSI segmentation, the extensive parameter requirements of these approaches hinder their deployment in resource-constrained mobile vision applications. Consequently, there is an urgent demand for lightweight models. Huang et al. [26] introduced a lightweight network that incorporates a squeeze-and-excitation module before feature fusion. Furthermore, Howard et al. [27] and Sandler et al. [28] presented MobileNets and MobileNetV2, which utilize deep separable convolution. However, the performance of these lightweight models on other datasets remains disappointing owing to their reliance on extremely large training data. Therefore, it is imperative to develop models that balance performance and lightness.

To address the above three issues, we introduce the stepped transmission optimization network (STONet) with knowledge distillation (KD) for model compression, denoting it as STONet-S* (STONet-S with KD). Our approach leverages knowledge learned from the teacher model (STONet-T) to enhance the accuracy of the student model (STONet-S). Within our STONet framework, we first propose a novel group enhancement and interaction unit (GEIU) with two variants that dynamically enhance the semantic representations of extracted features. Specifically, we construct group local and global enhancement mechanisms (GLGEMs) that adjust feature importance using importance coefficients generated for each group in both local and global feature space. Furthermore, two types of fusion strategies are established based on activation area distribution. Second, we implement a stepped transmission decoder (STD) comprising stepped capture modules (SCMs), self-reverse revision modules (SRRMs), and upsample (UP) modules. Considering STD, SCM adeptly captures valuable information from features at various levels, effectively mitigating interference. Simultaneously, SRRM meticulously learns precise residual details and revises features that have not been previously focused on, thereby preventing the loss of critical information. Additionally, we employ the UP module to seamlessly restore feature resolution, optimizing predictions incrementally and ensuring robustness.

Regarding our KD strategy, we recognize that significant features related to objects and backgrounds exhibit distinct patterns in the frequency domain. To address this, we adopt frequency-awareness KD (FAKD), which utilizes the discrete cosine transform (DCT) and octave convolution to transmit frequency domain knowledge from the teacher to the student. Furthermore, detail delivery KD (DDKD) assists the student in extracting and learning finer-grained information, and stepped-response KD (SRKD) addresses the diminishing ability of the student to represent features, which impacts its discrimination capability. To assess the generalizability of our STONet-S*, we conducted experiments on two publicly available datasets, achieving exceptional performance. The key contributions of this study can be succinctly summarized as follows:

1) In the proposed method, the GEIU enhances the extracted features and seamlessly integrates them, leveraging the inherent characteristics of multiscale features. In the decoding stage, we introduce the concept of STD to synergize with SCM and SRRM. This collaborative approach facilitates the capture of multiscale information from the bottom up. Consequently, our model avoids the loss of critical information and produces robust predictions.
2) Considering knowledge transfer from STONet-T to STONet-S, we propose three novel mechanisms: FAKD, DDKD, and SRKD. These mechanisms ensure that knowledge is effectively transferred without compromising performance while achieving model compression. Notably, FAKD, in particular, is tailored to encapsulate both high- and low-frequency knowledge within the frequency domain.
3) We conducted empirical investigations using the proposed STONet-S* on two publicly available datasets, comparing it with 23 state-of-the-art methods. Our findings unequivocally highlight the advantages inherent in our model.

II. Related Work

A. Semantic Segmentation

Recent years have witnessed significant progress in semantic segmentation methodologies. Shelhamer et al. [8] pioneered the use of fully convolutional networks (FCNs) with skip connections, enabling pixel-wise prediction. In addition, Badrinarayanan et al. [1] introduced an innovative upsampling process leveraging encoder pooling. To enhance semantic understanding, Chen et al. [29] introduced atrous spatial pyramid pooling (ASPP) in models such as DeeplabV2, DeepLabV3 [30], and DeepLabV3+ [31], effectively expanding the receptive field and abstracting essential features. Furthermore, Li et al. [32] proposed a novel kernel-based attention mechanism for fine-tuning extracted feature maps. Last, Ma et al. [33] leveraged foreground activation modules to model object features, achieving precise segmentation results.

Despite significant progress documented in these studies, complex scenes continue to pose segmentation challenges. The emergence of multimodal fusion techniques effectively addresses this predicament. For instance, Jiang et al. [34] proposed an encoder–decoder framework called RedNet, incorporating a residual module with skip connections to mitigate model degradation. Meanwhile, Zhou et al. [18] adeptly harnessed diverse backbones to extract information from two modalities via self-attention mechanisms. Seichter et al. [35] leveraged a ResNet-like structure [36] to establish a valid RGB-D segmentation method, serving as an initial processing step to enhance subsequent performance. Additionally, Zhang et al. [37] explored a cross-modal fusion for RGB-X semantic segmentation framework, called CMX, which achieves cross-modal fusion using both RGB and complementary modalities. Notably, Yang et al. [21] delved into multilevel feature cues, employing gating mechanisms to coordinate semantic extraction. Similarly, Zhou et al. [19] introduced a co-attention fusion module, enabling interaction between features from two modalities and feeding them into a hybrid branching specializer for prediction. Likewise, Liang et al. [20] devised a fusion strategy based on explicit attention, ensuring the extraction of complementary cross-modal features.

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While these methodologies effectively leverage multimodal information to enhance semantic segmentation accuracy, their direct applicability to RSIs remains limited. The specificity of RSIs—such as variations in shooting angles, target counts, and color texture—poses unique challenges. In contrast, our proposed STONet-S* demonstrates applicability to RSI tasks, yielding satisfactory results through effective multimodal fusion.

B. RSI-Oriented Semantic Segmentation

Semantic segmentation in RSI faces challenges owing to modality discrepancies and substantial intraclass variation. Existing approaches address this issue diversely. Xu et al. [38] introduced HRCNet for high-resolution RSIs, preserving semantic information without compromising spatial structure. Zhou et al. [39] iteratively improved cross-modal integration via a refined post-fusion module. Liu et al. [40] employed an adaptive fusion strategy to extract modality-specific semantic features. Wang et al. [41] proposed UNetFormer and FTUNetFormer to compare the performance of CNN-based and transformer-based methods on RSIs. Li et al. [42] developed a lightweight ABCNet, extracting RSI information through dual paths. In addition, Li et al. [43] enhanced learning by guiding feature fusion using attention mechanisms.

Although these models fully fuse features, they often overlook the variability in quality across RSIs. Effective utilization of the extracted features significantly impacts the final segmentation performance. Audebert et al. [44] explored the effect of the model on unbalanced quality during different fusion stages, revealing that late fusion can rectify errors from ambiguous data, while early fusion exhibits higher sensitivity. Li et al. [45] introduced a geometry-aware-based model that leverages 3-D elevation information to alleviate constraints arising from missing depth maps or depth map mismatches. Chen et al. [46] proposed edge perception for noise reduction in each modal image, enhancing feature fusion. In contrast, our approach embeds GLGEM within GEIU, allowing autonomous enhancement of feature representation by adjusting the importance of each group through significance coefficients in both global and local feature spaces.

While existing RSI-oriented segmentation methods perform satisfactorily, they overlook the challenge of balancing model computation difficulties with lightweight design. Some lightweight models sacrifice performance for parameter reduction. To address this, researchers are increasingly exploring KD, which significantly reduces model size without substantial accuracy degradation.

C. Frequency Domain Deep Learning

Several studies have focused on the frequency domain. Gueguen et al. [47] directly applied DCT to an image as input for a deep network in subsequent vision tasks. Ehrlich and Davis [48] introduced a transmission algorithm to convert spatial-domain features into frequency-domain features. To incorporate frequency-domain information into CNN models, Zhong et al. [49] proposed a novel frequency-enhancement module (FEM) that leverages a floating-point DCT followed by learnable enhancement. Lin et al. [50] designed a frequency-domain CNN architecture using chunk decomposition and unsaturated activation functions.

In contrast to existing approaches, we employ octave convolution [51] for online learning of frequency-domain features. Notably, our innovative use of high- and low-frequency information from octave convolution enhances knowledge transfer, allowing students to mimic more effective feature representations from the teacher network, bridging the gap between them.

D. KD

KD methods enable lightweight models without compromising accuracy. Chang et al. [52] investigated a matching paradigm with asymptotic stage-by-stage feature distillation. Zhang et al. [53] proposed a many-to-one method connecting the many layers of the teacher to a specific layer of the student. Yang et al. [54] transformed features into refined structure pixel-category correlation for knowledge transfer. Shan [55] introduced a novel pixel-by-pixel feature similarity approach to address extraction challenges. Li et al. [56] designed an instances-awareness KD strategy, guiding student networks to study both features and relationships. Zhang et al. [57] proposed a hierarchical KD method involving feature extraction, pixel-level distillation, and overall distillation. Wang et al. [58] effectively guided students using two different teacher networks for recognition through feature studying and category calibration.

Notably, KD has received extensive attention regarding RSIs. Chen et al. [59] employed a straightforward response-based distillation to align student model outputs with those of the teacher model. Zhang et al. [60] used teacher networks to fully exploit crucial foreground and background regions in RSIs, enabling student networks to replicate these multiscale features. In addition, Song et al. [61] introduced a novel relationship-based KD loss that effectively compresses models.

In most existing approaches, the architecture comprises a teacher with a large backbone, a student with a lightweight backbone, and associated distillation modules. These modules are modified primarily regarding channels and sizes. Inspired by this framework, we applied KD techniques to semantic segmentation in RSIs. Unlike prior studies, we introduced a novel FAKD to enable STONet-S to learn both high- and low-frequency knowledge from STONet-T. Additionally, we employed DDKD to enhance the encoder ability of the STONet-S to capture fine-grained features. Furthermore, we adopted SRKD to improve cross-level discrimination in STONet-S.

III. Proposed Method

A. Overall Architecture

In Fig. 1, we utilize ResNet-50 [36] as the backbone for STONet-T, and MobileNetV2 [28] for STONet-S. To tailor MobileNetV2 for semantic segmentation, we exclude the global average pooling layer and the last fully connected layer. Features R_i and D_i, corresponding to feature levels $i \in \{1, 2, 3, 4, 5\}$, result from processing infrared, red, and green (IRRG)

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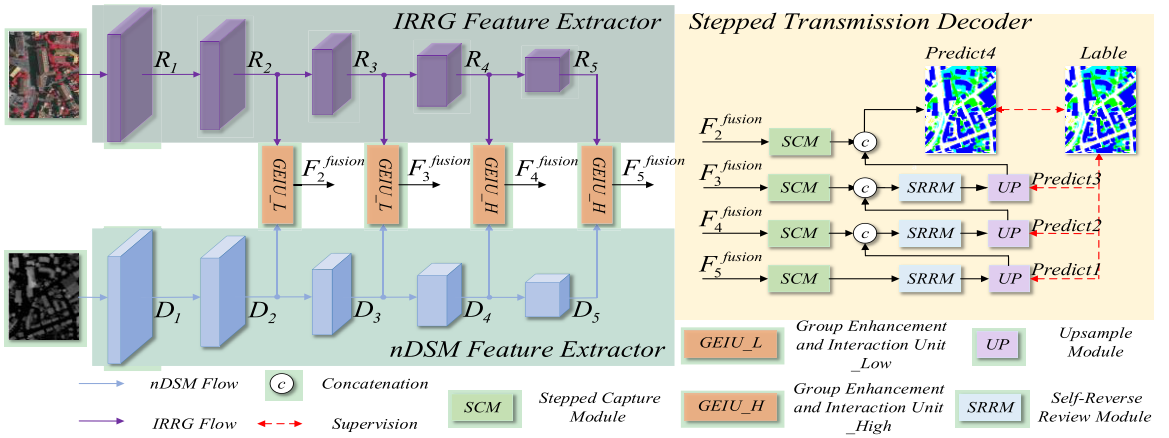


Fig. 1. Overview of our STONet-T/STONet-S overall framework. After extracting the low- and high-stage features from RSIs by ResNet- 50, we propose new GEIU_L and GEIU_H to fuse them, respectively. The output of GEIU is passed to the STD comprising two key modules (SCM and SRRM) to capture temporal information from the bottom up, learn accurate residual details, highlight salient target regions, and review features ignored in the past.

images and normalized digital surface model (nDSM) images using respective extractors. Despite the extensive information on first-level features, they also introduce interference [62]. Therefore, we selectively utilize the last four levels during the fusion stage. Specifically, low-level features (R_2, R_3, and D_2, D_3) are directed to CEIU_L, while high-level features (R_4, R_5,and D_4, D_5) enhance representation in CEIU_H, leveraging multiscale characteristics. The GEIU F_i^{fusion} output feeds into STD through four branches. In each decoding branch, we incorporate SCM and SRRM to facilitate multi-scale feature learning, thereby preserving critical information. Finally, the UP block interpolates the prediction map to meet task requirements. This type of stepped optimization decoder enables robust multiscale feature aggregation for segmentation mappings.

B. GEIU

Extracting representative information from features for semantic segmentation is crucial and can be achieved using the proposed GEIU. GEIU comprises two components: GEIU-L and GEIU-H. We create expressive feature representations by selectively aggregating hierarchical and distinct feature portions. These activations are often disrupted by similar patterns and complex backgrounds, which lead to inaccurate localization and recognition. To address this, we introduce GLGEM (see Fig. 2), which generates importance coefficients for each group. These coefficients significantly enhance the local and global distribution of semantic features, strengthen autonomous feature learning, and suppress noise. GLGEM is lightweight and requires minimal parameters or computations. Fig. 3 illustrates GEIU-L and GEIU-H.

1) Enhancement Part: In GLGEM, as depicted in Fig. 2, two branches exist: one for group local enhancement and the other for group global enhancement. Before aggregating the last four-level features, they undergo a 1×1 convolutionwith batch normalization and a leaky ReLU function. This process reduces channel count, thereby effectively minimizing computational cost.

In the group global enhancement branch, each reduced input is initially divided into G groups ($G = 4$ for sizes and number

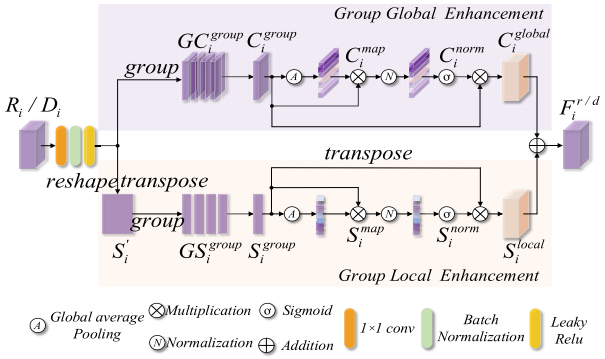


Fig. 2. Proposed GLGEM. The input is the feature extracted by the encoder.

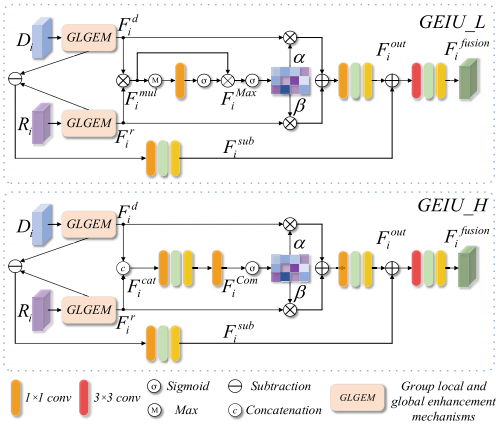


Fig. 3. Two parts of GEIU. The one above is for low-level features, and the one below is for high-level features. Before a comprehensive modal fusion is performed on the input information, the input features are fed into GLGEMs for enhancement.

of channels) along the channel dimension. Subsequently, each successively selected group is represented by a vector at every location in the space, denoted as $C_i^{\text{group}} = \{r_{1,\ldots,m}(r_i \in R^{C/G}, m = HW)$. The unavoidable noise and similar patterns often complicate the acquisition of well-distributed feature representations. To solve this problem, C_i^{group} is subjected to global average pooling and multiplied by itself to obtain C_i^{map}

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as follows:

$$C_i^{\text{group}} = \text{Group}(R_i / D_i) \tag{1}$$

$$C_i^{\text{map}} = C_i^{\text{group}} \times \text{Gap}\left(C_i^{\text{group}}\right) \tag{2}$$

where Group, Gap, and $\times$ represent the group operation, the global-average pooling operation, and element-wise multiplication, respectively. We use the scaling factors in the Group Normalization (GN) [63] to generate the corresponding importance coefficients $C_i^{\text{norm}} \in R^{B\times G\times H\times W}$. Specifically, we normalize C_i^{map} by subtracting the mean μ_c and dividing by the standard deviation σ_c, and the normalized features are fed into a sigmoid function as follows:

$$C_i^{\text{norm}} = \text{Sig}\left(w_c \frac{C_i^{\text{map}} - \mu_c}{\sigma_c + \varepsilon} + b_c\right) \tag{3}$$

where $\varepsilon(\varepsilon = 1e-5)$ is a constant and $+$ denotes element-wise addition. μ_c and σ_c are the mean and standard deviation of C_i^{map}, $w_c \in R^{1\times G\times 1\times 1}$ and $b_c \in R^{1\times G\times 1\times 1}$ are the trainable weights and bias of the affine transformation, whose values are initialized to 0 and 1, respectively. In contrast to the extensive parameter count across the entire network, this contribution remains almost negligible. Finally, to obtain the enhanced feature C_i^{global}, the original C_i^{group} is multiplied by the importance coefficients

$$C_i^{\text{global}} = C_i^{\text{group}} \times C_i^{\text{norm}}. \tag{4}$$

In the group local enhancement branch, operations remain consistent, except for features grouping along the height and width dimensions. Initially, we reshape and transpose the input into $S_i' \in R^{B\times HW\times C}$. Subsequently, we apply the Group operation to S_i' to group features along the HW dimension. Next, we compute the importance coefficients $S_i^{\text{norm}} \in R^{B\times G^2\times C}$ using the same approach

$$S_i' = \text{Trans}(\text{Shape}(R_i / D_i)) \tag{5}$$

$$S_i^{\text{group}} = \text{Group}(S_i') \tag{6}$$

$$S_i^{\text{map}} = S_i^{\text{group}} \times \text{Gap}(S_i^{\text{group}}) \tag{7}$$

$$S_i^{\text{norm}} = \text{Sig}\left(w_s \frac{S_i^{\text{map}} - \mu_s}{\sigma_s + \varepsilon} + b_s\right) \tag{8}$$

where Trans and Shape represent the transpose and reshape operations, $w_s \in R^{1\times G^2\times 1}$ and $b_s \in R^{1\times G^2\times 1}$, whose initial values are 0 and 1, respectively. Finally, we obtain the enhanced feature S_i^{local}

$$S_i^{\text{local}} = S_i^{\text{group}} \times S_i^{\text{norm}}. \tag{9}$$

The eventual output of GLGEM is the sum of the results of the two branches

$$F_i^{r/d} = C_i^{\text{global}} + S_i^{\text{local}}. \tag{10}$$

The GLGEM-enhanced features promptly propagate to the interaction part.

2) Interaction Part: Considering the four-level extracted features, the activation areas of the lower-level features exhibit extensive dispersion, whereas the areas corresponding to higher-level features demonstrate greater focus. As depicted in Fig. 3, to enhance interaction based on the aforementioned distinctiveness, we propose two interaction parts: GEIU-L and GEIU-H.

In GEIU-L, we focus on distinguishing mutual activation areas across two different modalities. To achieve this, we propose enabling low-level features to interact. Initially, we perform subtraction and apply a convolutional layer to the absolute values of F_i^d and F_i^r, resulting in F_i^{sub}. This enriched feature captures information about the differences between the two modalities. Next, we multiply F_i^d by F_i^r to identify common activation regions. For effective interaction among distributed features, we emphasize the informative region using a Max operation. Specifically, we employ a 1×1 convolution with a sigmoid function to generate a feature map, which we apply to F_i^{mul} via a residual connection. Subsequently, we use another sigmoid function to obtain attention graphs α and β. Finally, to obtain the fusion feature $F_i^{\text{fusion}} \in R^{C\times H\times W}$, we separately weight F_i^d and F_i^r with α and β, combine them with the difference map F_i^{sub}, and apply a 3×3 convolutional layer to restore the original size. Formally

$$F_i^{\text{sub}} = \text{CBL}\left(|\text{Sub}\left(F_i^d, F_i^r\right)|\right) \tag{11}$$

$$F_i^{\max} = \text{Sig}\left(C(\max)\left(F_i^d \times F_i^r\right)\right) \times \left(F_i^d \times F_i^r\right) \tag{12}$$

$$\alpha = \text{Sig}\left(F_i^{\max}\right), \quad \beta = 1 - \alpha \tag{13}$$

$$F_i^{\text{fusion}} = C_{3\times 3}\text{BL}\left(\text{CBL}\left(\left(F_i^d \times \alpha\right) + \left(F_i^r \times \beta\right)\right) + F_i^{\text{sub}}\right) \tag{14}$$

where Sub and Max are the subtraction and max operations, respectively; || denotes the absolute value; CBL denotes 1×1 convolution, batch normalization, and the leaky ReLU function; and C denotes 1×1 convolution while $C_{3\times 3}$ denotes 3×3 convolution. In high-level feature maps, semantic features exhibit variation across both modalities. Therefore, in the GEIU-H approach, we leverage the significant complementary features from distinct focused areas. Our process closely resembles the previous method, with the exception that we concatenate feature vectors F_i^d and F_i^r. Additionally, we introduce a 1×1 convolutional layer to reduce the channel count by half. This modification is justified because these features are already concentrated

$$F_i^{\text{Max}} = \text{Sig}\left(C\left(\text{CBL}\left(\text{Concat}\left(F_i^d, F_i^r\right)\right)\right)\right) \tag{15}$$

$$F_i^{\text{fusion}} = C_{3\times 3}\text{RL}\left(\text{CBL}\left(\left(F_i^d \times \alpha\right) + \left(F_i^r \times \beta\right)\right) + F_i^{\text{sub}}\right). \tag{16}$$

C. STD

In Fig. 1, STD is constructed during the decoding stage. Fusion features F_i^{fusion} are transferred from the bottom to the top within each branch, where we employ SCM and SRRM for multiscale feature learning and information recall. The UP block interpolates the prediction map to meet task requirements. This stepped refinement decoder enables multiscale feature aggregation for robust segmentation.

1) SCM: SCM (see Fig. 4) enhances multiscale information capture. It employs four global-average pooling operations with sizes of 6, 3, 2, and 1, and four 1×1 convolutions as residual connections. Given an input F_2^{fusion}, four 1×1 convolutions are adopted for transformation as follows:

$$F_2^2, F_2^3, F_2^4, F_2^5 = C\left(F_2^{\text{fusion}}\right) \tag{17}$$

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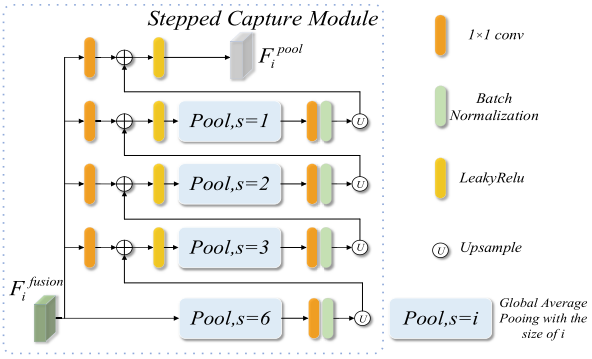


Fig. 4. Proposed SCM. The input is a fused feature.

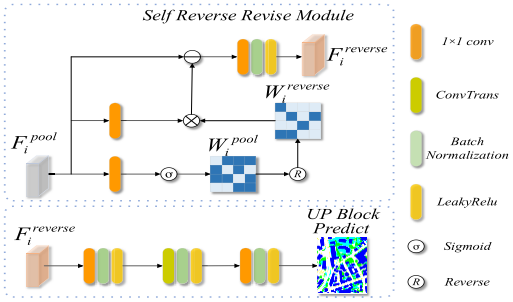


Fig. 5. Proposed SRRM and UP block.

where F_2^2, F_2^3, F_2^4, and F_2^5 represent the features injected into the four SCM branches from the bottom to the top. Subsequently, each feature is learned in the branch as follows:

$$F_2^{2'} = L\left(F_2^2 + \mathrm{Up}\left(\mathrm{CB}\left(\mathrm{Gap}_{\mathrm{size}=6}\left(F_2^{\mathrm{fusion}}\right)\right)\right)\right) \tag{18}$$

$$F_2^{3'} = L\left(F_2^3 + \mathrm{Up}\left(\mathrm{CB}\left(\mathrm{Gap}_{\mathrm{size}=3}\left(F_2^{2'}\right)\right)\right)\right) \tag{19}$$

$$F_2^{4'} = L\left(F_2^4 + \mathrm{Up}\left(\mathrm{CB}\left(\mathrm{Gap}_{\mathrm{size}=2}\left(F_2^{3'}\right)\right)\right)\right) \tag{20}$$

$$F_2^{\mathrm{pool}} = L\left(F_2^5 + \mathrm{Up}\left(\mathrm{CB}\left(\mathrm{Gap}_{\mathrm{size}=1}\left(F_2^{4'}\right)\right)\right)\right) \tag{21}$$

where $\mathrm{Gap}_{\mathrm{size}=i}$ is the global average pooling of size i, CB is the 1×1 convolution with batch normalization, Up is the bilinear upsampling operation, and L denotes the leaky ReLU function. Thus, SCM identifies features within receptive fields, progressively reducing their size. The SCM exhibits a step-shaped branch structure, where branches connect in pairs.

2) SRRM: While learning details for saliency refinement is intuitive, precise collection often requires additional operations. Inspired by the reverse attention mechanism [64], we propose an enhanced SRRM that prioritizes the feature itself. Rather than relying on weakly relevant information, our approach efficiently revises previously overlooked details.

In Fig. 5, the SRRM input is F_i^{pool}. We employ a 1×1 convolution with a sigmoid activation function to generate the weight map. Next, we apply a Reverse operation to obtain the reverse weight map, denoted as W_i^{reverse}. This map is then element-wise multiplied by the original feature. Finally, we compute the final output F_i^{reverse} by subtracting the feature map from the input and applying a 1×1 convolutional layer

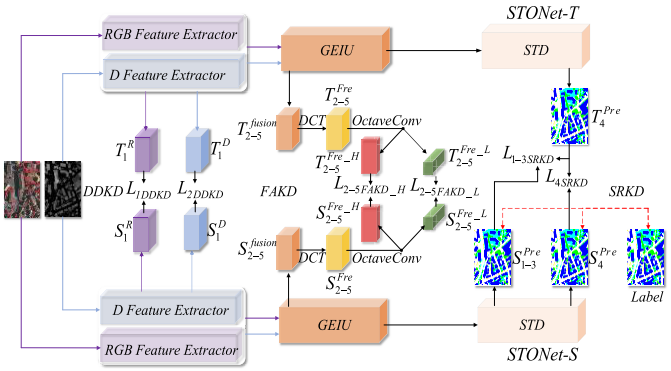


Fig. 6. Overall architecture of the proposed STONet-S*, comprising DDKD, FAKD, and SRKD.

for smoothing

$$W_i^{\mathrm{reverse}} = \mathrm{Rev}\left(\mathrm{Sig}\left(C\left(F_i^{\mathrm{pool}}\right)\right)\right) \tag{22}$$

$$F_i^{\mathrm{reverse}} = \mathrm{CBL}\left(\mathrm{Sub}\left(F_i^{\mathrm{pool}}, C\left(F_i^{\mathrm{pool}}\right) \times W_i^{\mathrm{reverse}}\right)\right) \tag{23}$$

where Rev denotes the reverse operation. Removing the predicted map from one side of the final output features guides the module to discover complementary information sequentially.

3) UP Block: Utilizing SCM and SRRM for the SRRM output. As depicted in Fig. 1, we generate task-appropriate feature maps, and to meet segmentation requirements, we propose a straightforward upsampling method (referred to as UP). UP comprises three small blocks. The ConvTrans block increases resolution, the first convolution block reduces channel count, and the last one doubles it. Both convolution and transposed convolution layers are followed by batch normalization and a leaky rectified linear unit function. The output channels match the number of segmentation classes.

D. STONet-S*

The integral architecture of the proposed STONet-S* is depicted in Fig. 6. STONet-S* comprises three key KDs: DDKD, FAKD, and SRKD. FAKD facilitates the study of feature extraction and interaction processes between STONet-S and STONet-T in the frequency domain. DDKD was proposed to enhance the capability of STONet-S to extract details during the encoding phase, while SRKD was presented to enhance the segmentation ability of STONet-S by simulating the segmentation results of STONet-T.

1) FAKD: Most distillation approaches primarily transfer knowledge about relevant features to the student network. However, recognizing that the significant features of the object and background are more distinguishable in the frequency domain, we developed FAKD to transfer frequency domain knowledge from the teacher to the student. Specifically, we utilize DCT [47] to convert fused information into the frequency domain. Contrary to the cited methods [47], [48], we employ octave convolution [51] for online learning of frequency maps. These maps are split into low and high-frequency components. The former captures moderate transitions, often representing most of the object, while the latter focuses on sharp transitions, such as object edges. By leveraging these two frequency maps, we facilitate knowledge transfer between student and teacher networks.

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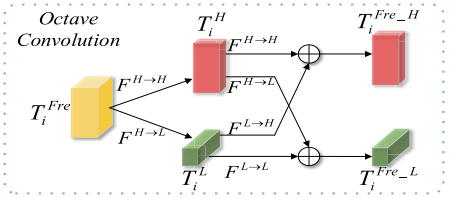


Fig. 7. Proposed octave convolution block.

The inputs to FAKD comprise fused features from STONet-T and STONet-S, denoted as T_i^{fusion} and S_i^{fusion}, respectively. T and S represent features originating from STONet-T and STONet-S. Given an input of T_i^{fusion}, we facilitate knowledge transfer by initially applying a 1×1 convolution with upsampling to standardize the resolutions of all inputs to $C \times H \times W$ ($C = 256$, $H = 32$, $W = 32$). Subsequently, we transform the fused features into the frequency domain using DCT to obtain $T_i^{\text{Fre}} \in R^{C\times H\times W}$. We then employ octave convolution to automatically capture both high and low-frequency information in an end-to-end manner. As depicted in Fig. 7, in this process, we explicitly factorize T_i^{Fre} along the channel dimension into $T_i^{\text{Frein}} = \{T_i^H, T_i^L\}$. The high-frequency maps $T_i^H \in R^{(1-\theta)C\times H\times W}$ (θ defaults to 0.75) capture fine details, while the low-frequency maps $T_i^L \in R^{\theta C\times 2H\times 2W}$ exhibit slower spatial variations. Octave convolution effectively mitigates congestion caused by DCT, thereby leveraging the computational speed of GPUs. The output of the octave convolution, denoted as $T_i^{\text{Freout}} = \{T_i^{\text{Fre_H}}, T_i^{\text{Fre_L}}\}$, can be succinctly expressed as

$$T_i^{\text{Fre_H}} = F\left(T_i^H; W^{H\to H}\right) + \text{Up}\left(F\left(T_i^L; W^{L\to H}\right), 2\right) \quad (24)$$

$$T_i^{\text{Fre_L}} = F\left(T_i^L; W^{L\to L}\right) + F\left(\text{pool}\left(T_i^H, 2\right); W^{H\to H}\right) \quad (25)$$

where $F(T; W)$ is a convolution with learnable parameters of W, pool(T, K) is an average pooling with a kernel size of $K \times K$, and Up(T, S) is a UP operation by a factor of S via bilinear interpolation.

Finally, FAKD uses the HCL function [65] to calculate the losses as

$$L_{\text{FAKD}} = \sum_{i=2}^{5} L_{\text{HCL}}\left(T_i^{\text{Fre_H}}, S_i^{\text{Fre_H}}\right) + L_{\text{HCL}}\left(T_i^{\text{Fre_H}}, S_i^{\text{Fre_H}}\right) \quad (26)$$

$$L_{\text{HCL}}(f_1, f_2) = \sum_{i=0}^{3} \frac{1}{2^i}\text{MSE}(f_1, f_2) + \sum_{l=1,2,4} \frac{\text{MSE}(\text{Aap}(f_1), \text{Aap}(f_2))}{2^l} \quad (27)$$

where MSE is the mean squared error loss, Aap denotes adaptive average pooling, and $l = 1$, 2, and 4 denotes the scale of adaptive average pooling.

2) DDKD: FAKD efficiently bridges the gap between student and teacher information extraction and interaction. We leverage extensive details from shallow layers, allowing the students to learn first-level features from the STONet-T feature extractor to enable detailed information extraction. In Fig. 7, we employ a convolutional layer to standardize the feature channels to 64. By combining IRRG and nDSM features, we drive STONet-S to replicate details from STONet-T using FAKD

$$L_{\text{DDKD}} = L_{\text{HCL}}\left(T_1^R, S_1^R\right) + L_{\text{HCL}}\left(T_1^D, S_1^D\right). \quad (28)$$

3) SRKD: To facilitate more comprehensive student learning, we propose SRKD. The SRKD framework comprises two main components (see Fig. 7). In the initial stage, we refine the student's final segmentation map S_4^{Fre} by leveraging the teacher's final segmentation map T_4^{Fre}. We employ the cross-entropy loss, consistent with the previously mentioned supervision method, to guide this correction. The equations are as follows:

$$L_{\text{4SRKD}} = L_{\text{CE}}\left(T_4^{\text{Fre}}, S_4^{\text{Fre}}\right) \quad (29)$$

$$L_{\text{CE}} = -\sum_{L}^{k} Y_i \times \log(P_i). \quad (30)$$

Next, we analyze the remaining three-layer prediction map generated by the network decoder outputs from both the teacher and student. Leveraging our STD gradual information correction and optimization from the bottom up, we employ cross-level supervision. We select the final prediction map generated by the teacher models to convey forecasting knowledge to the student models. In this approach, we employ the Kullback–Leibler divergence with the Dice loss to calculate losses as follows:

$$L_{\text{1-3SRKD}} = \sum_{i=1}^{3} L_{\text{KLD}}\left(T_i^{\text{Fre}}, S_i^{\text{Fre}}\right) + \sum_{i=1}^{3} L_{\text{Dice}}\left(T_i^{\text{Fre}}, S_i^{\text{Fre}}\right) \quad (31)$$

$$L_{\text{KLD}}\left(T_i^{\text{Fre}}, S_i^{\text{Fre}}\right) = \sum_{j=1}^{HW} D_{\text{KL}}\left(\text{soft}\left(\text{T}_\text{i}^{\text{Fre}}\right)||\text{soft}\left(\text{S}_\text{i}^{\text{Fre}}\right)\right) \quad (32)$$

where soft denotes the softmax function. Thus, the overall loss of SRKD is

$$L_{\text{SRKD}} = L_{\text{1-3SRKD}} + L_{\text{4SRKD}}. \quad (33)$$

E. Total Loss

In training STONet-S and STONet-T, we adopt the same multiple supervising approaches. We label the four-layer outputs of STD for monitoring. Specifically, the four outputs of STONet-S are denoted as S_n^{pre} ($n = 1$, 2, 3, 4). To compute pixel-level supervision for segmentation loss, we introduce cross-entropy

$$L_{\text{pre}} = \sum_{n=1}^{4} L_{\text{CE}}\left(S_n^{\text{pre}}, \text{Label}\right). \quad (34)$$

After training STONet-T, we distill the student network STONet-S by updating its parameters. STONet-S is trained using the total loss as follows:

$$L_{\text{total}} = L_{\text{pre}} + \delta(L_{\text{FAKD}} + L_{\text{DDKD}} + L_{\text{SRKD}}) \quad (35)$$

where δ is a hyper-parameter for balancing distillation loss. Distillation losses L_{FAKD} and L_{DDKD} are computed from intermediate layers, while L_{SRKD} is derived directly from segmentation maps. These principles can be readily applied in other image-processing research.

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TABLE I

PERFORMANCE COMPARISON BASED ON VAIHINGEN DATASET. THE BEST AND SECOND-BEST RESULTS IN EACH COLUMN ARE HIGHLIGHTED IN RED AND GREEN

	Year	Modal	Imp.surf		Building		Low veg.		Tree		Car		mAcc	mIoU
			Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU		
FCN[8]	2015	Single	89.66	79.71	93.22	86.80	75.83	64.33	89.22	75.58	45.12	40.16	78.61	69.32
U-Net[15]	2015	Dual	91.68	80.90	89.84	86.50	77.97	65.91	91.30	77.86	75.80	71.22	79.75	71.34
SegNet[1]	2017	Single	89.88	80.93	90.88	86.54	78.66	64.07	88.96	75.96	43.93	43.16	78.46	70.13
RedNet[34]	2018	Dual	91.49	84.62	94.81	91.07	78.67	66.59	91.41	78.27	59.77	56.06	83.23	75.32
DeepLabV3+[31]	2018	Single	90.06	81.11	87.04	82.7	76.65	64.44	88.60	76.64	42.51	43.10	76.97	69.49
ACNet[22]	2019	Dual	91.95	85.34	95.45	91.82	78.64	66.87	91.20	78.55	83.12	76.81	88.07	79.88
TSNet[18]	2020	Dual	87.93	78.98	95.81	91.47	71.62	57.03	94.26	81.26	67.63	66.86	83.54	75.12
SA-Gate[9]	2020	Dual	90.99	85.70	93.85	91.72	84.95	68.68	89.06	79.15	84.27	78.07	88.62	80.67
ESANet[35]	2021	Dual	92.09	85.18	94.93	91.16	75.72	65.48	92.35	77.65	75.92	70.11	86.20	77.92
HRCNet[38]	2021	Single	91.62	81.60	91.72	88.01	79.24	67.38	90.55	78.58	70.69	68.73	84.76	76.86
MANet[32]	2022	Single	91.57	83.23	93.77	90.04	76.29	65.59	91.91	77.94	86.03	81.01	87.91	79.56
CMANet[10]	2022	Dual	92.02	85.06	95.29	91.15	77.35	66.06	91.48	78.54	73.56	68.89	85.94	77.94
CMX[37]	2022	Dual	91.78	85.39	95.15	91.28	77.00	66.29	92.59	78.85	83.23	78.58	87.95	80.08
UNetFormer[41]	2022	Single	91.74	85.26	95.36	91.89	78.69	67.33	91.89	78.63	82.98	79.30	88.13	80.48
FTUNetFormer[41]	2022	Single	89.42	78.02	89.34	83.88	70.78	60.15	91.12	74.09	53.42	50.73	78.82	69.38
MGCNet[21]	2022	Dual	91.18	84.89	95.58	91.52	76.22	65.74	92.76	78.53	79.56	75.33	87.06	79.20
ABCNet[42]	2022	Single	86.60	80.49	93.55	87.87	80.77	63.72	87.24	74.16	18.14	17.90	73.26	64.83
SFAF-MA[24]	2023	Dual	90.94	85.37	95.67	92.21	80.99	67.55	90.78	78.95	74.74	71.19	86.62	79.05
A2-FPN[43]	2022	Single	92.12	82.54	93.64	90.59	77.36	65.58	90.36	78.82	89.82	81.04	88.80	79.61
CATNet[25]	2024	Dual	90.53	83.98	95.27	91.08	78.27	65.62	90.57	77.72	70.28	61.85	84.98	76.05
CEKD[12]	2023	Dual	92.22	85.05	95.78	91.66	76.57	66.77	92.69	78.97	76.63	72.01	86.71	78.89
EAEFNet[20]	2023	Dual	93.78	84.37	95.35	92.25	79.87	67.98	89.11	79.00	81.54	74.24	87.93	79.57
CPNet[11]	2024	Dual	91.79	78.58	93.99	88.97	61.96	55.58	90.77	74.31	29.63	27.68	74.23	65.02
STONet-T	2024	Dual	92.21	85.94	96.05	92.95	79.58	68.72	92.29	79.87	90.91	86.35	90.21	82.77
STONet-S	2024	Dual	91.82	84.73	95.29	91.69	78.93	67.14	90.83	78.23	74.63	68.47	86.30	78.05
STONet-S*	2024	Dual	92.13	85.51	95.66	92.85	78.99	67.71	91.83	79.10	86.45	81.80	89.01	81.40

IV. EXPERIMENTS

A. Datasets and Evaluation Metrics

1) Datasets: We evaluated the performance of the proposed STONet-S* on two publicly available datasets from the International Society for Photogrammetry and Remote Sensing: the Vaihingen and Potsdam datasets. Both datasets cover six categories: impermeable surfaces, buildings, low vegetation, trees, cars, and clusters/backgrounds. The Vaihingen dataset incorporates 16 fully annotated images, the equivalent three-channel IRRG, and single-channel nDSM images, with each pixel corresponding to a ground truth distance of 9 cm. The resolution of the images is 2500 × 2500 pixels. We divided the dataset into two portions: 11 images for training and 5 images (ID11, 15, 28, 30, and 34) for validation. The Potsdam dataset comprises 24 fully annotated images, the equivalent five-channel IRRG, and the single-channel nDSM images, at a resolution of 6000 × 6000 pixels. Each pixel corresponds to a ground truth distance of 5 cm. We selected seven images (IDs 2_11, 2_12, 4_10, 5_11, 6_7, 7_8, and 7_10) for validation and the remainder were used as the inputs during the training period.

2) Evaluation Metrics: We adopted various metrics to evaluate our STONet-T*. For each category, we used the class accuracy (Acc) and intersection over union (IoU). For overall performance, we employed the mean accuracy of each class (mAcc) and the mean IoU (mIoU).

B. Implementation Details

We employed the Pytorch platform for our experiments with STONet-S* on an NVIDIA TITAN XP GPU with CUDA 11.0 and 12 GB memory. The input and output were unified as 256 × 256 pixels through a sliding window. The label and input were enhanced by random scaling and cropping operations. Other transformations such as counterclockwise rotations of 90°, 180°, and 270°, as well as vertical and horizontal flipping, were employed. The proposed STONet framework consisted of two training periods. During the first period, we trained STONet-T based on the ResNet-50 backbone and determined its weights. During the second period, the optimization of STONet-S was supervised by the segmentation loss L_{pre} as well as the knowledge transfer losses L_{FAKD}, L_{DDKD}, and L_{SRKD}. We trained STONet-S for 120 epochs using the Adam optimizer with a weight decay of 0.0005, a batch size of 16, and an initial learning rate of 1×10^{-4}. We set the momentum parameter to 0.9, which decreases to 0.1 times its original value after every 20 epochs. The G in GLGEM was 4, δ in (35) was experimentally set to 1.

C. Comparisons With SOTA Methods

We compared STONet-S* with the following 23 state-of-the-art approaches: FCN [8], U-Net [15], SegNet [1], RedNet [34], DeepLabV3+ [31], ACNet [22], TSNet [18], SA-Gate [9], ESANet [35], HRCNet [38], MANet [32], CMANet [10], CMX [37], UNetFormer [41], FTUNetFormer [41], MGCNet [21], ABCNet [42], SFAF-MA [24], A2-FPN [43], CATNet [25], CEKD [12], EAEFNet [20], and CPNet [11]. Tables I and II present the segmentation results.

Notably, RSIs have various complexities and specificities. Despite the distinguishing features between object categories, pixel features can be easily blurred in complicated scenes, especially where objects are intertwined. As illustrated in Fig. 8, the similarity of pixel features between low vegetation (cyan) and trees (green), as well as the fact that the trees can occlude other pixels and low vegetation in their vicinity, often leads to pixel-feature confusion. Our STONet is slightly less effective in low vegetation and tree categories, which is an area where our model needs improvement.

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TABLE II

PERFORMANCE COMPARISON OF THE POTSDAM DATASET. THE BEST AND SECOND-BEST RESULTS IN EACH COLUMN ARE HIGHLIGHTED IN RED AND GREEN

	Year	Modal	Imp.surf		Building		Low veg.		Tree		Car		Cluter		mAcc	mIoU
			Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU		
FCN[8]	2015	Single	89.47	79.77	90.69	83.60	85.13	71.12	82.86	71.23	91.02	81.53	49.05	36.49	78.61	69.32
U-Net[15]	2015	Dual	90.03	80.27	88.71	82.92	85.82	71.60	84.06	72.05	93.89	90.24	50.30	36.26	82.13	72.22
SegNet[1]	2017	Single	90.18	80.46	90.21	84.18	85.88	71.63	82.49	70.68	93.15	89.72	51.76	37.21	82.28	72.31
RedNet[34]	2018	Dual	92.19	82.83	93.61	90.13	87.00	73.22	83.00	71.77	93.36	90.08	56.74	43.51	84.32	75.26
DeepLabV3+[31]	2018	Single	91.57	82.49	91.78	87.59	87.36	73.63	85.45	73.32	93.89	90.04	53.80	43.54	83.97	75.10
ACNet[22]	2019	Dual	91.32	82.74	93.83	90.06	86.16	73.53	86.03	72.87	93.79	90.43	54.51	41.65	84.27	75.21
TSNet[18]	2020	Dual	85.22	76.85	91.85	86.65	88.52	67.98	78.75	67.49	78.22	76.85	37.49	30.85	76.68	67.78
SA-Gate[9]	2020	Dual	85.84	80.64	93.65	88.51	86.46	72.71	85.70	72.89	92.18	89.39	62.70	40.59	84.42	74.12
ESANet[35]	2021	Dual	91.38	82.92	93.69	89.82	87.10	73.16	82.48	70.81	93.08	88.53	55.68	43.38	83.90	74.77
HRCNet[38]	2021	Single	90.03	81.68	90.87	85.75	88.17	73.18	82.02	71.32	93.94	89.82	56.72	40.03	83.63	73.63
MANet[32]	2022	Single	90.48	81.20	87.58	83.47	88.78	72.74	82.69	71.54	90.96	88.73	46.16	38.11	81.11	72.63
CMANet[10]	2022	Dual	91.75	81.87	93.64	90.23	86.42	73.59	84.70	72.40	92.74	89.11	55.97	42.74	84.20	74.99
CMX[37]	2022	Dual	90.54	81.86	93.80	89.28	87.17	73.11	83.52	71.45	92.23	87.99	51.61	40.58	83.14	74.04
UNetFormer[41]	2022	Single	91.07	81.76	91.87	87.05	84.84	72.87	86.83	73.18	94.81	89.42	55.27	41.05	84.11	74.22
FTUNetFormer[41]	2022	Single	87.77	77.93	87.98	81.38	86.68	68.23	72.06	59.72	92.40	87.52	44.98	35.54	78.65	68.39
MGCNet[21]	2022	Dual	90.61	81.21	93.87	89.02	86.72	72.67	83.70	72.51	93.39	88.91	53.24	39.66	83.59	74.00
ABCNet[42]	2022	Single	91.77	77.86	89.20	81.75	87.62	72.02	78.31	70.09	91.91	87.74	31.89	29.70	78.45	69.86
SFAF-MA[24]	2023	Dual	90.63	82.06	93.31	89.67	85.76	72.35	83.51	70.95	92.55	88.34	60.03	42.40	84.30	74.30
A2-FPN[43]	2022	Single	90.05	82.49	90.82	85.52	86.69	73.29	85.99	73.36	94.54	90.26	55.28	41.72	83.89	74.49
CATNet[25]	2024	Dual	88.12	79.80	93.32	89.52	85.91	70.66	78.78	65.85	91.56	84.99	55.58	40.41	82.21	71.87
CEKD[12]	2023	Dual	91.09	80.89	93.37	88.97	84.70	71.71	85.18	71.62	93.01	89.35	52.63	38.18	83.33	73.45
EAEFNet[20]	2023	Dual	92.32	82.04	93.22	89.54	84.34	72.37	85.02	71.91	94.64	88.30	53.55	39.34	83.85	73.92
CPNet[11]	2024	Dual	89.36	75.08	87.59	85.02	85.99	67.01	73.81	60.45	39.93	38.42	36.54	25.91	68.87	58.65
STONet-T	2024	Dual	92.08	83.42	93.77	90.37	87.27	74.50	85.38	73.94	95.51	92.25	59.75	46.45	85.63	76.82
STONet-S	2024	Dual	91.01	81.91	93.26	89.52	89.05	72.86	80.52	71.65	92.75	89.76	52.73	42.63	83.24	74.22
STONet-S*	2024	Dual	91.01	83.04	93.25	90.18	87.35	73.82	85.00	73.51	95.04	91.47	61.85	45.01	85.58	76.17

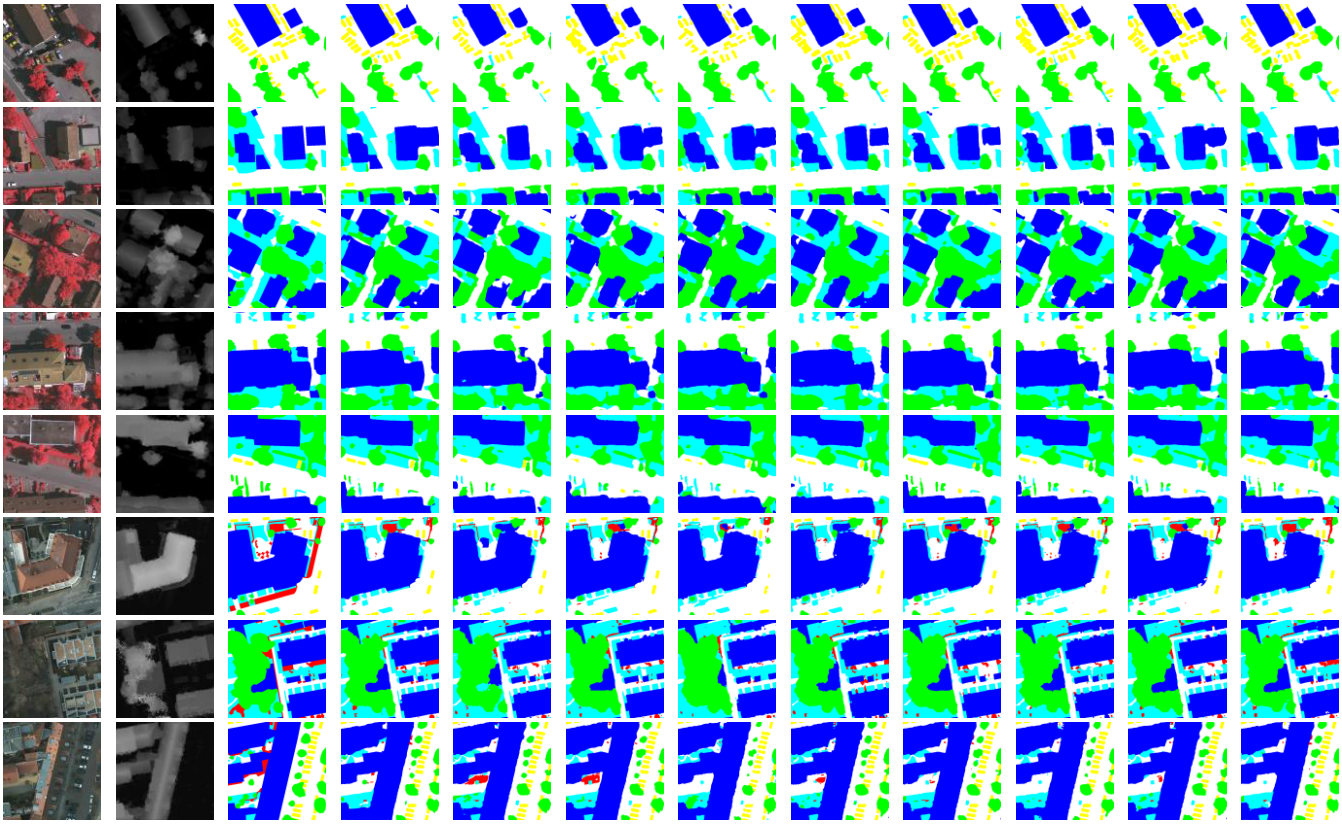


Fig. 8. Segmentation prediction results. (a) RGB, (b) nDSM, (c) Label, (d) STONet-T, (e) HRCNet, (f) RedNet, (g) TSNet, (h) SA-Gate, (i) ACNet, (j) CMX, (k) CMANet, and (l) STONet-T*.

1) Quantitative Evaluation: The quantitative results are summarized in Tables I and II, including the performance in each category and overall. Our proposed STONet-T and STONet-S contained the same modules except for the backbone. Overall, STONet-T achieved excellent results in all metrics, having an improvement of 1.59% and 2.10% in terms of mAcc and mIoU in the Vaihingen dataset and 1.21% and 1.72% in the Potsdam dataset. In the first dataset, our method

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achieved nine first-place and two second-place results; in the second dataset, the corresponding numbers were nine and one. As shown in Tables I and II, the performance of STONet-S (without KD) is still very far from optimal. However, the quantitative results of the proposed STONet-S* show a huge improvement effect from our distillation strategy. In the first dataset, STONet-S* achieved an improvement of 2.71% and 3.35% in terms of mAcc and mIoU; simultaneously, our proposed KD methods achieved almost 13% and 12% in terms of Acc and IoU in the car category. In the second dataset, STONet-S* achieved an improvement of 2.34% and 1.95% in terms of mAcc and mIoU: the student reached almost the same segmentation level as the teacher.

2) Qualitative Evaluation: Fig. 9 shows the segmentation prediction results of some models mentioned above and our design to provide a more comprehensive demonstration of the benefits of STONet-S*. We selected five sets of images from the Vaihingen dataset and three groups from the Potsdam dataset as controls. The results indicate that our method performed segmentation on the two datasets accurately and effectively in multiple categories. Our proposed STONet-S* method was skilled enough to split buildings (rows 2, 3, 4, 7, and 8), impervious surfaces, low vegetation, and trees (rows 3, 4, 5, 6, 7, and 8) from the background. It performed particularly well in identifying small cars (rows 1, 5, 6, and 8). Although it could not reach the same high level as the label in terms of cluster segmentation, it was markedly superior to other networks. We also observed that the proposed STONet-T and STONet-S* exhibited relatively similar performance, further verifying the practicality of the proposed KD framework.

3) Complexity Comparison: To demonstrate the advantages of our KD strategy in lightweighting, we used two metrics to compare the complexity of STONet-S* with that of other state-of-the-art models: total modal parameters (Params) and floating-point operations per second (FLOPS), as shown in Table III. Our STONet-T used ResNet-50 as the backbone to effectively mine multimodal features with few parameters and maintain relatively low FLOPS. By contrast, our STONet-S adopted MobileNetV2, enhancing its lightweightness. Our proposed method outperformed the current optimal model. Thus, we obtained 7.19 M for Params and 1.91 Gmac for FLOPS. Furthermore, our model had a 35.0 M weight size and an FPS of 41.2 frames/s. The results reveal that our STONet-T* is competitive and effective.

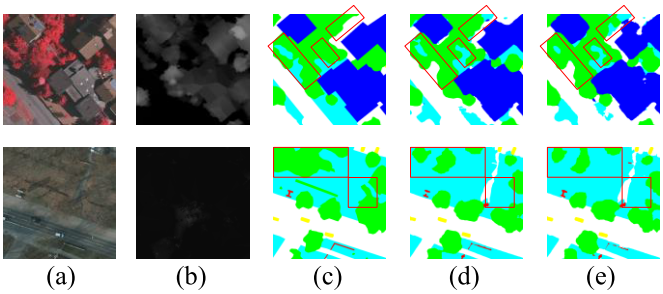


Fig. 9. Visualization images of the problems existing in RSIs from the Vaihingen and Potsdam datasets. (a) IRRG, (b) nDSM, (c) label, (d) STONet-T, and (e) STONet-S*. For perspective, we have highlighted crucial prediction areas using red boxes.

TABLE III
COMPLEXITY COMPARISON

Model	Year	Encoder	Params(M)	Flops(Gmac)
FCN[8]	2015	VGG-16	134.29	74.55
U-Net[15]	2015	VGG-16	26.36	55.93
SegNet[1]	2017	VGG-16	53.56	18.3
RedNet[34]	2018	ResNet-50	81.95	21.17
DeepLabV3+[31]	2018	ResNet-101	59.33	32.45
ACNet[22]	2019	ResNet-50	116.6	26.41
TSNet[18]	2020	ResNet-34/VGG-16	41.8	34.27
SA-Gate[9]	2020	ResNet-101	110.85	41.23
ESANet[35]	2021	ResNet-34	45.42	10.15
HRCNet[38]	2021	ResNet-50	62.71	30.27
MANet[32]	2022	ResNet-50	35.86	19.49
CMANet[10]	2022	ResNet-50	117.82	28.61
CMX[37]	2022	SegFormer	66.56	14.26
UNetFormer[41]	2022	ResNet-18	2.93	11.68
FTUNetFormer[41]	2022	Swin-Transformer	36.17	16.85
MGCNet[21]	2022	ConvNeXt	209.99	47.51
ABCNet[42]	2022	ResNet-18	13.67	3.91
SFAF-MA[24]	2023	ResNet-50	232.08	18.24
A2-FPN[43]	2022	ResNet-18	12.16	22.82
CATNet[25]	2024	Swin-Transformer	262.74	76.79
CEKD[12]	2023	ResNet-50	245.03	68.61
EAEFNet[20]	2023	ResNet-50	77.97	23.3
CPNet[11]	2024	Swin-Transformer	222.91	57.56
STONet-T	2024	ResNet-50	115.03	27.40
STONet-S	2024	MobileNetV2	7.19	1.91
STONet-S*	2024	MobileNetV2	7.19	1.91

D. Ablation Study

To verify the effectiveness of each significant component in STONet-S*, we performed comprehensive ablation studies using a control variable method with the modules. These studies were categorized into two types: those measuring the contribution of the modules in STONet-S* and those measuring the usefulness of its distilled portion. Table IV shows the effectiveness of our module in the two datasets with KD. Table V shows the impact of the two branches (global and local) of the GEIU module—the symbol $\surd$ indicates that the corresponding module was used. Table VI lists the contrasting effects of using only GEIU_L or only GEIU_H. Tables VII and VIII list that our KD strategies can effectively transmit knowledge from STONet-T to the lightweight STONet-S from the w/o (without) and addiction perspectives, respectively. Finally, Table IX lists the impact of different hyper-parameters on the final distillation effect.

1) Effectiveness of GEIU Module: To evaluate the effects of GEIU, we used three ablation methods. Tables IV–VI show the settings and related results. In the first verification experiment, we removed all modules and structures from our designs, marked them as the backbone, and used the element-wise addition operation instead of GEIU on the extracted features as the fusion. In the decoding part, we only used UP, concatenation, and simple convolution to generate predictions. Based on the backbone, each module was iterated separately. When only the backbone was used, the relative performance was poor despite the use of KD. Table IV and Fig. 10 reveal that adding the GEIU module resulted in several added details (more complete outlines of cars and low vegetation). Our GEIU module effectively integrated features of different sizes and eliminated interference. For instance, on Vaihingen, the mAcc and mIoU increased by 3.46% and 3.59%, respectively, while Potsdam increased by 1.91% and 1.49%. In Table V, it is evident that the segmentation effect was not as good when using only the global or local branch

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TABLE IV
ABLATION EXPERIMENTS FOR EACH MODULE OF STONET-S*

backbone	GEIU	SCM	SRRM	V		P	
				mAcc	mIoU	mAcc	mIoU
√				82.65	74.86	81.96	71.45
√	√			86.11	78.45	83.87	72.94
√	√	√		87.79	80.29	84.34	74.36
√	√	√	√	89.01	81.40	85.58	76.17

TABLE V
ABLATION EXPERIMENTS FOR GLGEM MODULE OF STONET-S*

Group Global Enhance	Group Local Enhance	V		P	
		mAcc	mIoU	mAcc	mIoU
		86.54	78.37	84.34	74.41
√		88.05	80.42	84.56	74.85
	√	87.12	79.73	84.63	74.91
√	√	89.01	81.40	85.58	76.17

TABLE VI
ABLATION EXPERIMENTS FOR GEIU MODULE OF STONET-S*

GEIU_H	GEIU_L	V		P	
		mAcc	mIoU	mAcc	mIoU
√		88.37	80.87	85.25	75.82
	√	88.30	80.72	84.65	75.28
√	√	89.01	81.40	85.58	76.17

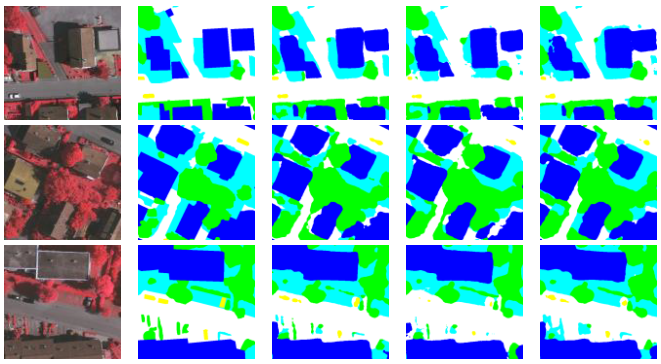


Fig. 10. Visualization images of the effectiveness of GEIU. (a) IRRG, (b) label, (c) STONet-S*, (d) backbone, and (e) backbone with GEIU.

of GLGEM compared to both combined. In contrast to no module, they could better correct for noise interference and highlight more salient regions before performing cross-modal feature fusion. In Table VI, when using only GEIU_L or GEIU_H in the fusion phase, we found that the performance of semantic segmentation decreased compared to using GEIU_L for low-level features and GEIU_H for high-level features. This implies that the design of the specific interactions based on different feature characteristics was effective.

2) Effectiveness of SCM Module: The proposed SCM module focuses on better capturing temporal multiscale information. Fig. 11 presents the ablation results with and without SCM. With SCM, our model performed better on segmenting fragmented cars, complexly distributed low vegetation, and impermeable surfaces. Table IV illustrates that applying SCM improved mAcc and mIoU by 1.68% and 1.84%, respectively, on the Vaihingen dataset. A similar trend was observed on the Potsdam dataset, where mAcc and mIoU achieved a rise of 0.47% and 1.42%, respectively. The validity of the stepwise

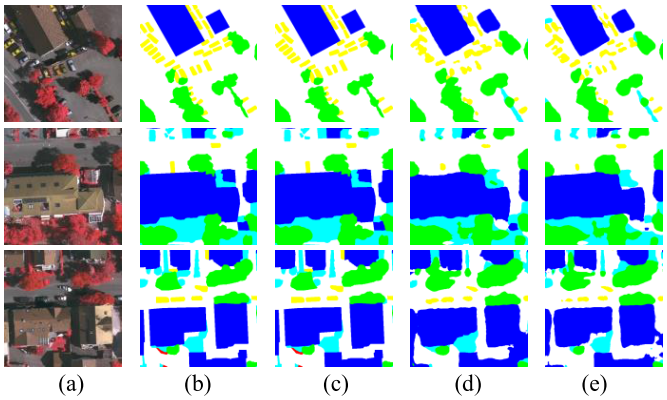


Fig. 11. Visualization images of the effectiveness of SCM. (a) IRRG, (b) label, (c) STONet-S*, (d) backbone with GEIU, and (e) backbone with GEIU and SCM.

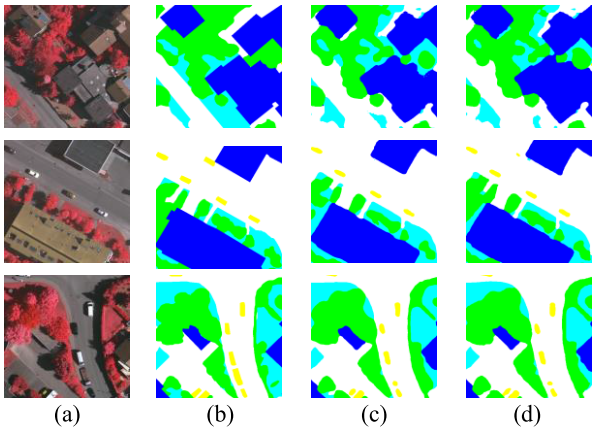


Fig. 12. Visualization images of the effectiveness of SSRM. (a) IRRG, (b) label, (c) backbone with GEIU and SCM, and (d) STONet-S*.

search for temporal changes from large to small acceptance fields was thus demonstrated.

3) Effectiveness of SRRM Module: We used SRRM to revise features by efficiently highlighting a salient target region and reviewing the features that were not previously focused on. Fig. 12 shows that this module allows the model to accomplish the semantic segmentation task effectively with a performance close to that of the label. Table IV illustrates that using SRRM increased mAcc and mIoU by 1.22% and 1.11%, respectively, on the Vaihingen dataset. The Potsdam dataset experienced an increase of 1.24% and 1.81%.

4) Effectiveness of KD Strategy: To evaluate the effectiveness of our KD strategy, we compared the distilled portions in STONet-S* from two perspectives, shown in Tables VII and VIII. Fig. 13 shows the gradual enhancement of the segmentation effect as we add our distillation method step by step. Results from both datasets reveal that our STONet-S* achieves performance close to that of STONet-T, particularly on the Potsdam dataset.

5) Effectiveness of Hyper-Parameters: The proposed KD introduces an additional hyper-parameter δ to control the ratio of different losses. Results from an ablation study are presented in Table IX. It is observed that when δ ranging from 0.1 to 1, our STONet-S* had an improvement of 0.91% and 0.95% in terms of mAcc and mIoU on the Vaihingen dataset and 2.18% and 1.76% on the Potsdam dataset, respectively, while it would drop moderately after sustained growth. In the

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TABLE VII
ABLATION EXPERIMENTS OF VARIOUS KD STRATEGIES USING W/O

	V		P	
	mAcc	mIoU	mAcc	mIoU
STONet-T	90.21	82.77	85.63	76.82
STONet-S	86.30	78.05	83.24	74.22
STONet-S*	89.01	81.40	85.58	76.17
w/o DDKD	87.97	80.19	84.89	75.13
w/o FAKD	88.27	80.71	84.76	74.99
w/o SRKD	88.06	80.47	84.78	75.37

TABLE VIII
ABLATION EXPERIMENTS OF VARIOUS KD STRATEGIES USING ADDICTION

DDKD	FAKD	SRKD	V		P	
			mAcc	mIoU	mAcc	mIoU
			86.30	78.05	83.24	74.22
√			87.36	79.43	83.92	74.77
√	√		88.06	80.47	84.78	75.37
√	√	√	89.01	81.40	85.58	76.17

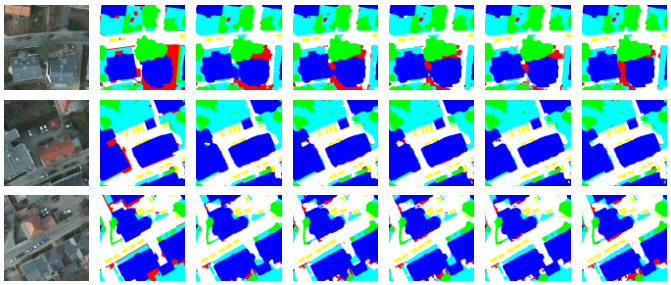


Fig. 13. Visualization images of the effectiveness of KD methods. (a) IRRG, (b) label, (c) with DDKD, (d) with DDKD and SRKD, (e) with DDKD and FAKD, (f) with FAKD and SRKD, and (g) STONet-S*.

TABLE IX
ABLATION EXPERIMENTS OF HYPER-PARAMETERS OF KD LOSS

δ	V		P	
	mAcc	mIoU	mAcc	mIoU
0	86.30	78.05	83.24	74.22
0.1	88.10	80.45	83.40	74.41
0.2	88.52	80.79	84.15	75.05
0.5	88.96	81.29	84.82	75.27
1	89.01	81.40	85.58	76.17
2	88.91	80.94	84.95	76.16
10	88.38	80.59	84.85	75.59

worst situation ($\delta = 0.1$), our KD still leads to an accuracy boost.

V. CONCLUSION

In this study, we propose a novel approach for semantic segmentation based on KD. GEIU from the fusion phase of our proposed STONet-S* enhances the extraction features and ensures full interaction. STD, which includes SCM and SRRM, steps up to capture multiscale information from the bottom up. In the KD strategy, we effectively leverage the ease of distinguishability of objects and backgrounds in the frequency domain and design FAKD, which transfers both high-frequency and low-frequency knowledge to STONet-S, facilitating the learning capacity of STONet-S. Supported by DDKD and SRKD, our model achieves good performance with a small number of parameters.Experimental results obtained on two datasets demonstrate the superiority of the proposed STONet-S*. Although we achieved good results with STONet-S*, further improvement is required in distinguishing between tree and low vegetation categories.

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